

Lecture 2: Air Pollution

- What is Environmental Pollution
- Definition of Air Pollution with examples.
- Classification of Air Pollutants
- Causes/Sources of Air Pollution or Air Pollutants
- Control/Preventive /Remedial Measures to Minimize Air Pollution

WHAT IS ENVIRONMENTAL POLLUTION

- For normal and healthy living a conducive environment is required by all living beings including humans, plants, micro-organisms and the wildlife.
- The favourable unpolluted environment has a specific composition. When this composition gets changed by addition of harmful substances, the environment is called polluted environment and the substances polluting it are called pollutants.
- ***Environmental pollution can, therefore be defined as any undesirable change in the physical, chemical or biological characteristics of any component of the environment (air, water, soil) which can cause harmful effects on various forms of life or property.***



DEFINITION OF AIR POLLUTION

- ***Air pollution is an atmospheric condition in which certain substances (including the normal constituents in excess) are present in concentrations which can cause undesirable effects on man and his environment.*** These substances include gases, particulate matter, radioactive substances etc.
- Gaseous pollutants: include oxides of sulphur (mostly SO_2 , SO_3), oxides of nitrogen (mostly NO and NO_2 and NO_x), carbon monoxide (CO), volatile organic compounds (mostly hydrocarbons) etc.

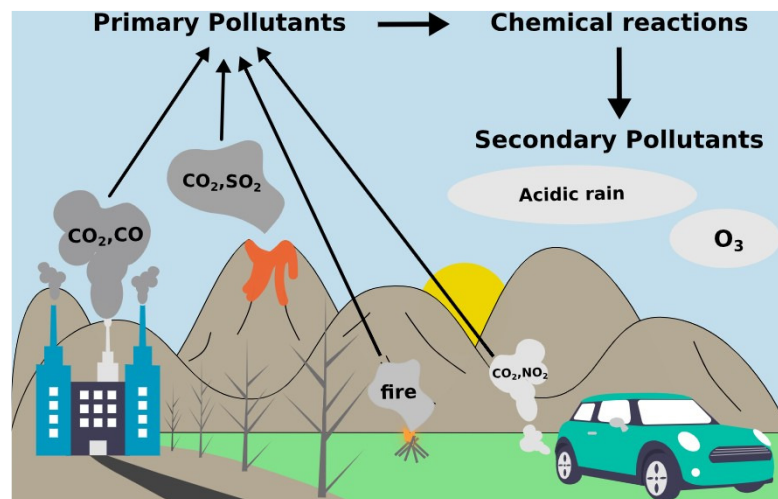
- Particulate pollutants: include smoke, dust, soot, fumes, aerosols, liquid droplets, pollen grains etc.
- Radioactive pollutants: include radon-222, iodine-131, strontium-90, plutonium-239 etc.

CLASSIFICATION OF AIR POLLUTANTS

- **On the basis of origin of pollutants**, air pollutants can be classified as primary or secondary pollutants.

(a) Primary Pollutants: These are emitted directly from the point source (identifiable source) e.g., carbon monoxide (CO), oxides of nitrogen (NO_x), oxides of sulphur (SO_x), hydrocarbons, radioactive substances etc.

(b) Secondary Pollutants: These are formed by interaction of primary pollutant (s) with other primary pollutant (s) or with some natural constituents of atmosphere, e.g., ozone (O_3), peroxyacetyl nitrate (PAN), photochemical smog etc.



CAUSES/SOURCES OF AIR POLLUTANTS

- Sources of air pollution are **Natural** and **Man-made (Anthropogenic)**.

- **Natural Sources:**

1. Volcanic Eruptions
2. Forest Fires
3. Sea Salt Sprays
4. Biological Decay
5. Photochemical Oxidation of Terpenes
6. Marshes
7. Extra Terrestrial Bodies

8. Pollen Grains of Flowers, spores etc.

9. Radioactive minerals present in the earth crust are the sources of radioactivity in the atmosphere.

- **Man-made Sources (Anthropogenic):**

1. *Thermal Power Plants:* Thermal power plants have become the major sources for generating electricity in India as the nuclear power plants couldn't be installed as planned. The main pollutants emitted are *fly ash and SO₂*.

2. *Industrial Units:* Industrial like nuclear power plants, iron and steel, metals and minerals, pulp and paper, tanneries, textile, chemical, foundry, cement, fertilizer, distilleries, pharmaceutical, mining etc. are of major concern. These industries discharge organic pollutants, particulates, gaseous pollutants, hydrocarbons, volatile organic compounds, hydrogen sulphide, lead, hazardous waste, oils and solvents etc.

3. *Vehicular Emissions:* Automobiles release gases such as carbon monoxide (about 77 %), oxides of nitrogen (about 8 %) and hydrocarbons (about 14 %). Heavy duty diesel vehicles spew more NO_x and suspended particulate matter (SPM) than petrol vehicles which produce more carbon monoxide and hydrocarbons.

4. *Fossil Fuel Burning:* Coal contains varying amounts of sulphur which on burning produces sulphur dioxide. Fossil fuel burning produces black soot. These pollutants i.e., CO, SO₂, soot and many others like formaldehyde, benzo- (a) pyrene (BAP) are toxic and harmful for health. BAP is also found in cigarette smoke and is considered to cause cancer.

5. *Agricultural Activities:* Agricultural air pollution comes mainly in the form of ammonia, which enters the air as a gas from heavily fertilized fields and livestock waste. It then combines with pollutants from combustion—mainly nitrogen oxides and sulphates from vehicles, power plants and industrial processes—to create tiny solid particles, or aerosols, no more than 2.5 micrometres across, about 1/30 the width of a human hair. The particles can penetrate deep into lungs, causing heart or pulmonary diseases.

CONTROL/PREVENTIVE/REMEDIAL MEASURES TO MINIMIZE AIR POLLUTION

- Siting of industries after proper Environmental Impact Assessment studies.
- Using low sulphur coal in industries.
- Removing sulphur from coal (by washing or with the help of bacteria).
- Removing NO_x during the combustion process.

- Vehicular pollution can be checked by regular tune-up of engines; replacement of more polluting old vehicles; installing catalytic converters; by engine modification to have fuel efficient (lean) mixtures to reduce CO and hydrocarbon emissions; and slow and cooler burning of fuels to reduce NOx emission (Honda Technology).
- Using mass transport system, bicycles etc.
- Shifting to less polluting (clean) fuels (hydrogen gas).
- Using non-conventional sources of energy.
- Using biological filters and bio-scrubbers.
- Planting more trees.
- Reduction of pollution at source.